

Negative Numbers in Binary				Giannacco
	Sign and Magnitude	1's Complement	2's Complement	
Description	The MSB is the sign 0: Positive 1: Negative	Flip every bit 0: Positive 1: Negative	Flip every bit and add one 0: Positive 1: Negative	
Example	$0110 = 6$ $1110 = -6$	$0110 = 6$ $1001 = -6$	$0110 = 6$ $1010 = -6$	
Special Cases	Two combinations for 0 $0000 = 0$ $1000 = 0$	Two combinations for 0 $0000 = 0$ $1111 = 0$	One combination for 0 $0000 = 0$ $0000 = -1$	
Notes	Simple to understand, but difficult to implement with hardware	More complex arithmetic	Easier arithmetic but overflow detection is needed	